

SEARCY, AR, USA

WMO: 722188

Lat: 35.212N

Lon: 91.737W

Elev: 81

StdP: 100.36

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.5	(c) -6.3	(d) -15.5	(e) 1.0	(f) -4.7	(g) -12.8	(h) 1.3	(i) -4.2	(j) 8.9	(k) 10.1	(l) 8.2	(m) 7.3	(n) 2.2	(o) 10	(p) 0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	36.8	25.3	35.3	25.6	34.0	25.2	27.6	33.6	26.9	32.8	26.3	32.0	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.7	30.2	25.4	20.8	29.7	24.8	20.1	29.1	87.6	33.6	84.8	33.0	82.1	31.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.3	6.3	-12.4	39.0	2.6	2.8	-14.3	41.0	-15.8	42.7	-17.2	44.3	-19.1	46.3		
			-13.1	28.6	2.4	0.6	-14.8	29.0	-16.2	29.3	-17.5	29.7	-19.3	30.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	3.9	6.3	11.0	16.2	21.0	26.1	27.3	26.8	23.6	16.9	10.1	5.6	(6)
	DBStd	9.48	5.81	6.01	5.49	4.23	3.97	2.51	2.56	2.81	3.33	4.72	5.01	5.20	(7)
	HDD10.0	618	204	135	53	5	1	0	0	0	0	4	59	156	(8)
	HDD18.3	1862	447	339	232	88	23	0	0	0	2	86	251	394	(9)
	CDD10.0	2914	16	31	86	191	342	484	537	522	407	218	62	20	(10)
	CDD18.3	1116	0	1	6	24	106	234	278	263	159	41	3	1	(11)
	CDH23.3	11537	0	7	56	228	947	2537	3092	2772	1475	399	23	1	(12)
	CDH26.7	5354	0	0	4	36	328	1237	1577	1385	664	121	2	0	(13)
(14) Wind	WSAvg	2.5	2.9	3.2	3.3	3.1	2.5	2.2	2.0	2.0	2.1	2.2	2.4	2.7	(14)
(15) Precipitation	PrecAvg	1250	102	93	129	128	130	73	89	67	89	113	108	123	(15)
	PrecMax	1962	186	203	251	358	286	162	200	139	243	358	260	222	(16)
	PrecMin	935	32	40	40	32	35	29	29	13	8	0	21	17	(17)
	PrecStd	269	47	39	56	74	63	31	43	38	55	88	58	53	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	24.5	27.2	29.5	33.2	37.8	39.2	39.2	36.2	32.3	26.6	21.9	(19)
		MCWB	17.2	18.8	17.9	21.4	23.4	24.9	24.8	26.6	24.3	22.5	19.2	17.7	(20)
	2%	DB	18.4	21.3	25.0	27.7	31.6	35.7	36.8	36.4	34.5	29.8	23.1	19.2	(21)
		MCWB	15.5	16.4	18.0	20.0	22.9	25.6	25.6	25.3	24.6	21.6	17.7	16.8	(22)
	5%	DB	16.7	19.0	22.5	26.1	30.2	34.1	35.1	34.7	32.9	27.7	20.9	16.9	(23)
		MCWB	13.6	15.1	16.1	19.0	22.7	25.3	26.0	25.3	24.7	20.3	16.8	13.8	(24)
	10%	DB	13.9	16.5	20.1	24.3	28.6	32.7	33.7	33.2	30.9	25.5	18.5	14.3	(25)
		MCWB	10.5	13.3	15.0	18.0	21.8	24.7	25.8	25.1	23.2	19.0	14.5	11.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.3	20.0	20.4	23.2	25.6	27.8	28.2	28.4	26.9	25.0	21.6	19.8	(27)
		MCDB	19.8	22.4	25.7	27.2	30.3	33.9	33.6	34.3	32.9	30.0	24.2	21.0	(28)
	2%	WB	16.9	18.1	19.1	21.4	24.5	26.9	27.6	27.6	25.9	23.6	19.4	17.4	(29)
		MCDB	18.1	20.4	23.3	25.9	29.2	32.9	33.5	33.8	32.0	27.7	21.7	18.8	(30)
	5%	WB	14.4	15.9	17.6	20.4	23.7	26.2	26.9	26.9	25.2	22.1	17.4	14.8	(31)
		MCDB	15.5	18.1	20.7	24.2	28.2	32.2	32.6	32.8	31.1	25.3	19.8	16.3	(32)
	10%	WB	10.8	13.4	16.1	19.3	22.8	25.5	26.2	26.3	24.5	20.7	15.3	11.9	(33)
		MCDB	12.8	15.7	19.2	22.7	27.0	31.1	31.8	31.6	29.2	23.5	17.9	13.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.8	11.7	12.4	11.6	11.6	11.5	11.5	12.2	12.9	12.1	10.3	(35)
		MCDBR	13.7	14.1	14.7	14.4	13.6	13.5	13.0	13.5	14.2	15.2	14.8	13.1	(36)
	5% WB	MCWBR	10.0	10.2	8.6	7.5	6.1	5.0	4.2	5.0	5.4	7.5	9.3	10.0	(37)
		MCWBR	10.1	11.4	12.4	11.3	11.4	11.7	11.6	11.7	12.4	12.1	11.2	11.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.321	0.329	0.362	0.399	0.433	0.463	0.493	0.482	0.415	0.360	0.331	0.316	(40)
		taud	2.478	2.454	2.393	2.286	2.253	2.183	2.142	2.152	2.329	2.459	2.494	2.507	(41)
	Ebn at Noon		881	915	908	886	856	827	799	801	843	868	862	866	(42)
		Edh at Noon	85	97	112	131	138	148	153	149	119	96	83	78	(43)
(44) All-Sky Solar Radiation	RadAvg		2.46	2.99	4.13	5.21	5.72	6.51	6.24	5.76	4.90	3.78	2.69	2.14	(44)
	RadStd		0.21	0.41	0.34	0.38	0.43	0.45	0.32	0.39	0.38	0.49	0.29	0.23	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page